

**CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY**

COLLEGE OF COMPUTER STUDIES

Software Design Description

for
playIT



Change History Signature

Version	Date	Description	Author
0.1	May 11, 2026	SDD based on SRS (v2)	Development Team
0.2	May 15,2026	Added word challenge(as advised by our adviser)	Dev Team
1.0	May 20,2026	Refactored Game Module: Decoupled the original Game Module into atomic components. Absorbed by (sayit, findit, blendit)	Dev Team

Preface

This Software Design Description document was developed by the playIT development team as part of the IT332-56 course project for the College of Computer Studies at Cebu Institute of Technology - University. The document translates the functional requirements defined in the Software Requirements Specification (SRS v2.0) into a concrete architectural and detailed design for the playIT application. playIT is a gamified English phonics mobile application designed specifically for Grade 1 Filipino learners, following the Department of Education's Marungko reading sequence.

Key design decisions documented in this SDD include:

- MVVM + Clean Architecture for separation of concerns
- Local Room/SQLite database for complete offline functionality
- Vosk offline speech recognition for privacy-preserving pronunciation feedback
- Candy Crush-style winding level map for engaging progression

This document is intended for the development team, course advisers, and project evaluators. It serves as the blueprint for implementation and maintenance of the playIT system.

The team acknowledges the guidance of our course adviser, whose feedback led to the inclusion of the Word Challenge (Blend It) module as a checkpoint after each letter group and the support for multiple user with its own progression per device.

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1. Introduction

1.1. Purpose

This Software Design Description document provides the architectural and detailed design specifications for playIT, a gamified English phonics mobile application for Grade 1 Filipino learners. This document translates the requirements defined in the SRS into a concrete design that guides implementation.

1.2. Scope

This SDD covers the complete design of playIT including:

- Android mobile application architecture using MVVM + Clean Architecture
- Local Room/SQLite database schema for all offline data persistence
- Offline speech recognition integration via Vosk library
- Jetpack Compose UI component design for all 10 screens
- Gamification engine (hearts, stars, streaks, badges)
- Candy Crush-style winding level map with 26 Marungko letter nodes
- Letter grouping system — 28 letters in n groups, each group ends with a Word Challenge
- Parent/Guardian offline progress dashboard and PDF report generation
- Multi-profile support — up to 6 child profiles per device, each with independent progress
- Word Challenge module — word building game unlocked after completing all letters in a group

1.3. Definitions and Acronyms

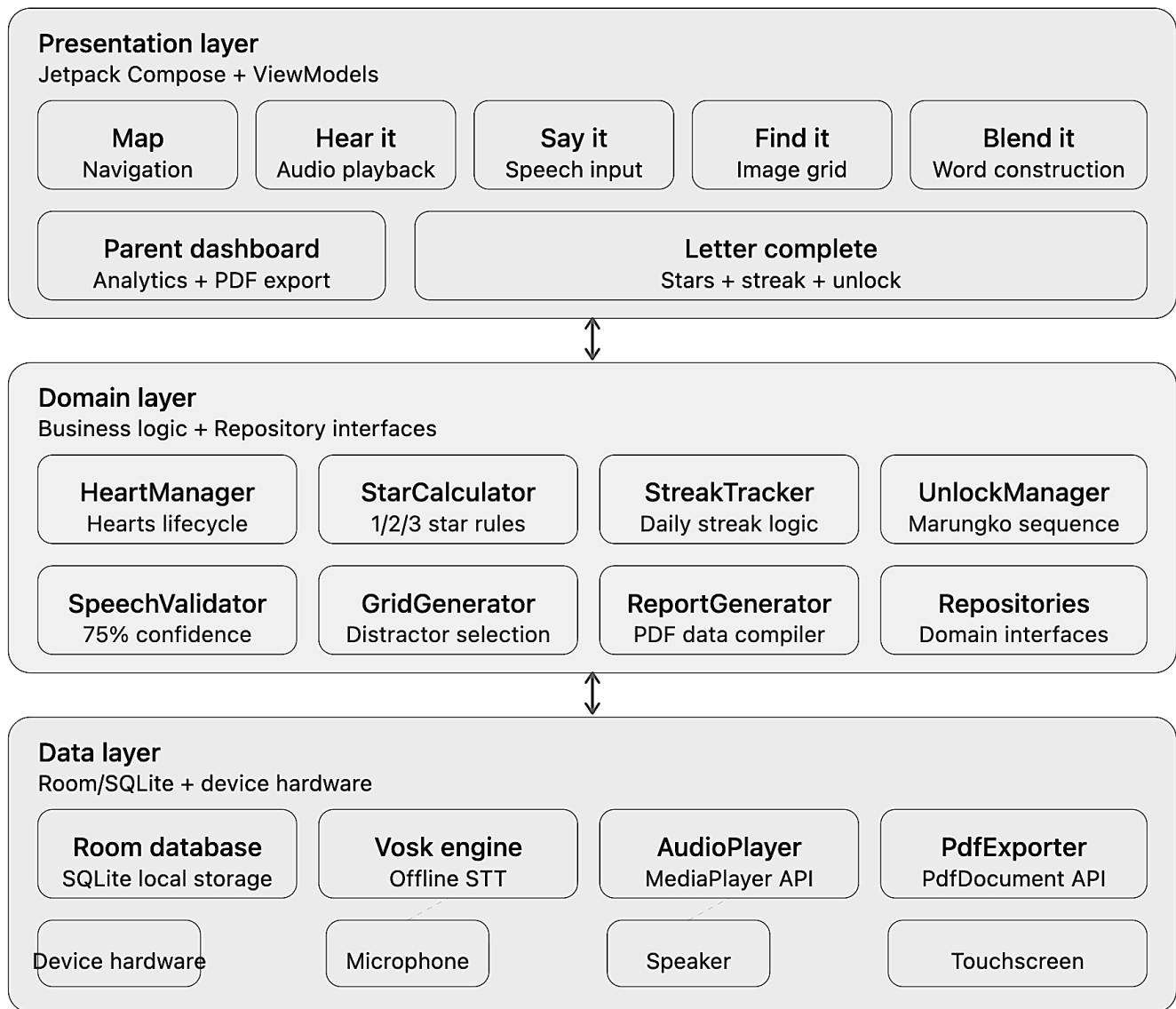
MVVM	Model-View-ViewModel — Android architecture pattern
Room	Android ORM library wrapping SQLite
Vosk	Offline speech recognition library for Android
Composable	A UI function in Jetpack Compose
ViewModel	MVVM component that holds and manages UI state
Repository	Abstraction layer between ViewModel and data source
DAO	Data Access Object — Room interface for database queries

Marungko	DepEd reading sequence of 28 letters by frequency of use
Phoneme	Distinct unit of sound in English
STT	Speech-to-Text
ERD	Entity Relationship Diagram

1.4. References

- playIT SRS v2.0 — Development Team
- Project Proposal REVISED 4 — Team 2526-sem2-it332-56
- IEEE 1016 SDD Standard
- Android Developer Documentation — developer.android.com
- Vosk Documentation — alphacephei.com/vosk
- DepEd Grade 1 English Curriculum Competencies

2. Architectural Design



playIT follows a Clean Architecture pattern with MVVM (Model-View-ViewModel) on the presentation layer. The system is divided into three main layers:

Layer	Purpose	Components
Presentation Layer	UI and user interaction	Jetpack Compose Screens, ViewModels
Domain Layer	Business logic and use cases	HeartManager, StarCalculator, StreakTracker, UnlockManager, Repository interfaces

Data Layer	Data persistence and external sources	Room Database, Vosk Recognizer, MediaPlayer, AudioCapture
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Technology Stack Summary

Layer	Technology	Version	Purpose
Language	Kotlin	1.9+	Primary development language
UI Framework	Jetpack Compose	1.5+	Declarative UI
Architecture	MVVM + Clean Architecture	—	Separation of concerns
Database	Room (SQLite)	2.6+	Local offline data persistence
Speech Recognition	Vosk	0.3.47	Offline phoneme detection
Audio Playback	Android MediaPlayer	API 26+	Pre-recorded phoneme audio
PDF Generation	Android PdfDocument API	API 26+	Local report export (no extra library)
Image Loading	Coil	2.5+	Async image loading from assets
Concurrency	Kotlin Coroutines + Flow	1.7+	Async operations
Minimum SDK	API 26 (Android 8.0)	—	Device compatibility
Navigation	Jetpack Navigation Compose	2.7+	Screen routing with phonemeld arguments

Screen Inventory

Screen	Purpose
SplashScreen	App loading — routes to ProfileSelectScreen on every launch
ProfileSelectScreen	Grid of existing child profiles — select a profile or create a new one
NamePromptScreen	Child name and avatar entry — shown only when creating a new profile

MapScreen	Candy Crush winding path map of 28 letter nodes and 7 Blend It checkpoint nodes, scoped to active profile
HearItScreen	Audio modeling for any phoneme — loads by phonemeld navigation argument
SayItScreen	Speech recognition for any phoneme — loads by phonemeld navigation argument
FindItScreen	Picture discrimination for any phoneme — loads by phonemeld navigation argument
LetterCompleteScreen	Star rating celebration shown after all 3 sublevels of a letter are complete
BlendItScreen	Guided word construction game — loads by groupId, unlocked after all 4 letters in group complete
BlendItCompleteScreen	Star drop-in celebration and session summary shown after Blend It session ends
ParentDashboardScreen	Offline progress dashboard — now profile-aware with profile switcher dropdown
ReportPreviewScreen	PDF content preview and save to local storage — scoped to selected profile

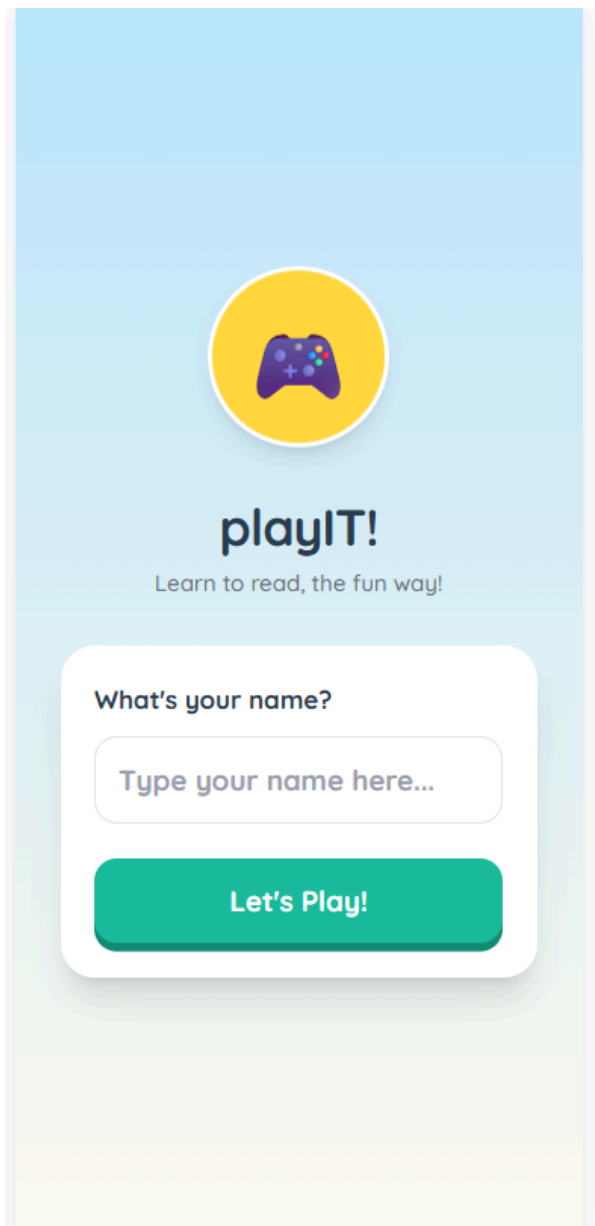
3. Detailed Design

Module 1 profile + Map and Navigation

1.1 Display Profile, Level Map & Navigate to Letters

- User Interface Design

First time open screen



mapscreen



- Front-end component(s)

Component	Description	Type
ProfileSelectScreen	Grid of profile cards shown on app start — select or create a profile	Screen
ProfileCard	Shows avatar, child name, last played date, total stars	Composable
AddProfileButton	Opens NamePromptScreen for new profile creation	Composable
NamePromptScreen	Child name + avatar entry for new profiles only	Screen
AvatarPicker	Grid of animal avatar options to choose from	Composable
MapScreen	Root composable, scrollable winding path container scoped to active profile	Screen
LetterNode	Individual letter bubble showing letter symbol, star count, lock state	Composable
BlendItChallengeNode	Distinct banner-shaped checkpoint node — locked until all 4 letters in group complete	Composable
PathConnector	Line segment connecting two nodes — variant style near Blend It checkpoints	Composable
MascotBubble	Animal mascot with contextual speech bubble	Composable
TopStatsBar	Streak flame + active profile name + total star count	Composable
StreakBadgeUnlock	Badge notification overlay shown on streak milestones	Composable

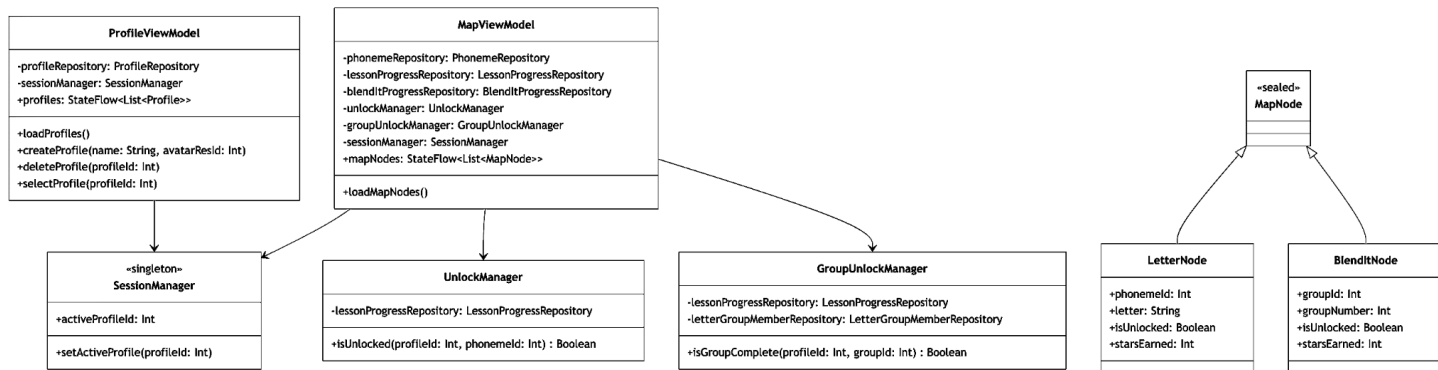
- **Back-end component(s)**

Component	Description	Type
ProfileViewModel	Loads all profiles, handles create and delete, sets active session	ViewModel
MapViewModel	Builds combined node list of letter nodes and Blend It nodes scoped to activeProfileId	ViewModel
ProfileRepository	CRUD operations on Profile table	Interface
ProfileRepositoryImpl	Room DAO implementation for Profile table	Class
PhonemeRepository	Fetches all 28 phonemes in Marungko sequence	Interface

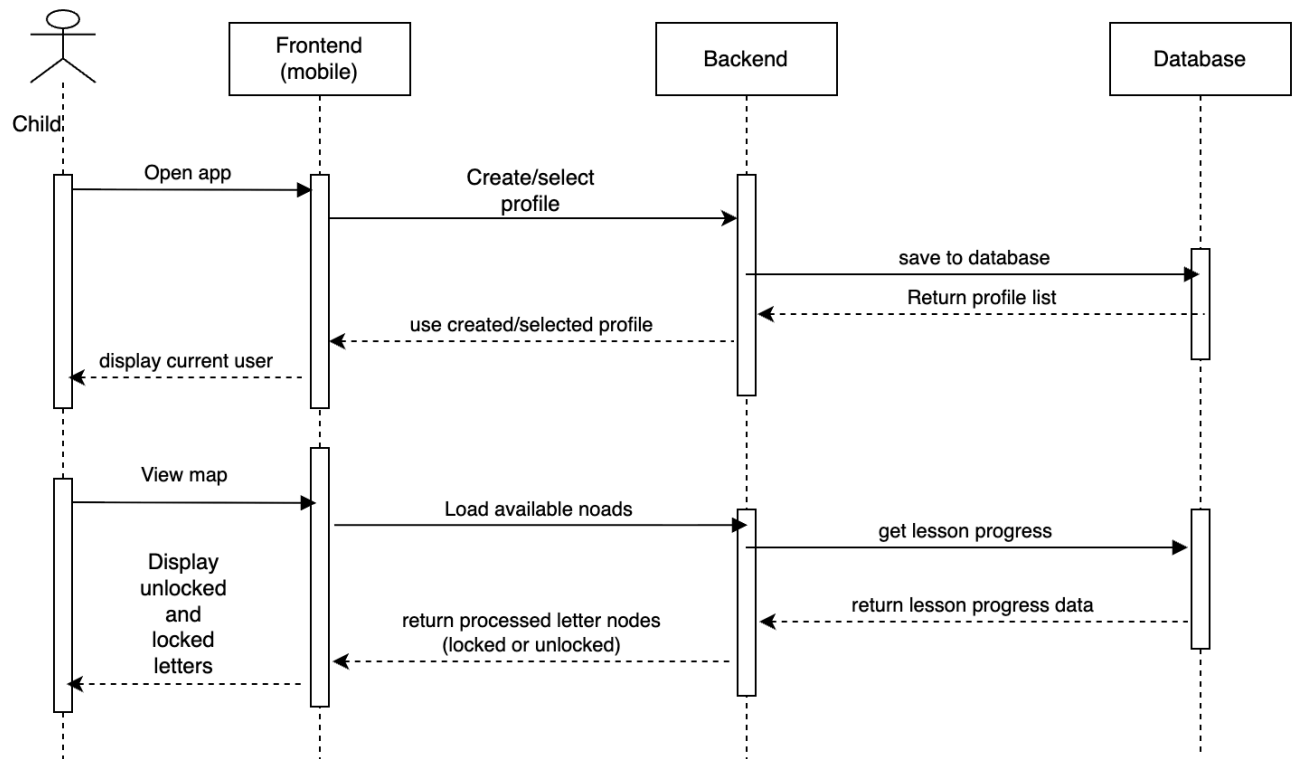
PhonemeRepositoryImpl	Room DAO — fetches Phoneme joined with LessonProgress by profileId	Class
LessonProgressRepository	Reads completion status per phoneme per profile for map unlock logic	Interface
BlendItProgressRepository	Reads Blend It completion status per group per profile for node unlock	Interface
UnlockManager	Checks if Letter N all 3 sublevels complete — unlocks Letter N+1	Class
GroupUnlockManager	Checks if all letters in a group complete via LetterGroupMember — unlocks Blend It node	Class
StreakTracker	Updates daily streak per profile, checks 5/10/15/20 milestones, resets after 24h	Class
SessionManager	In-memory singleton holding activeProfileId for the current session	Object
AchievementRepository	Flips isUnlocked on Achievement rows when milestone earned for active profile	Interface

- Object-Oriented Components

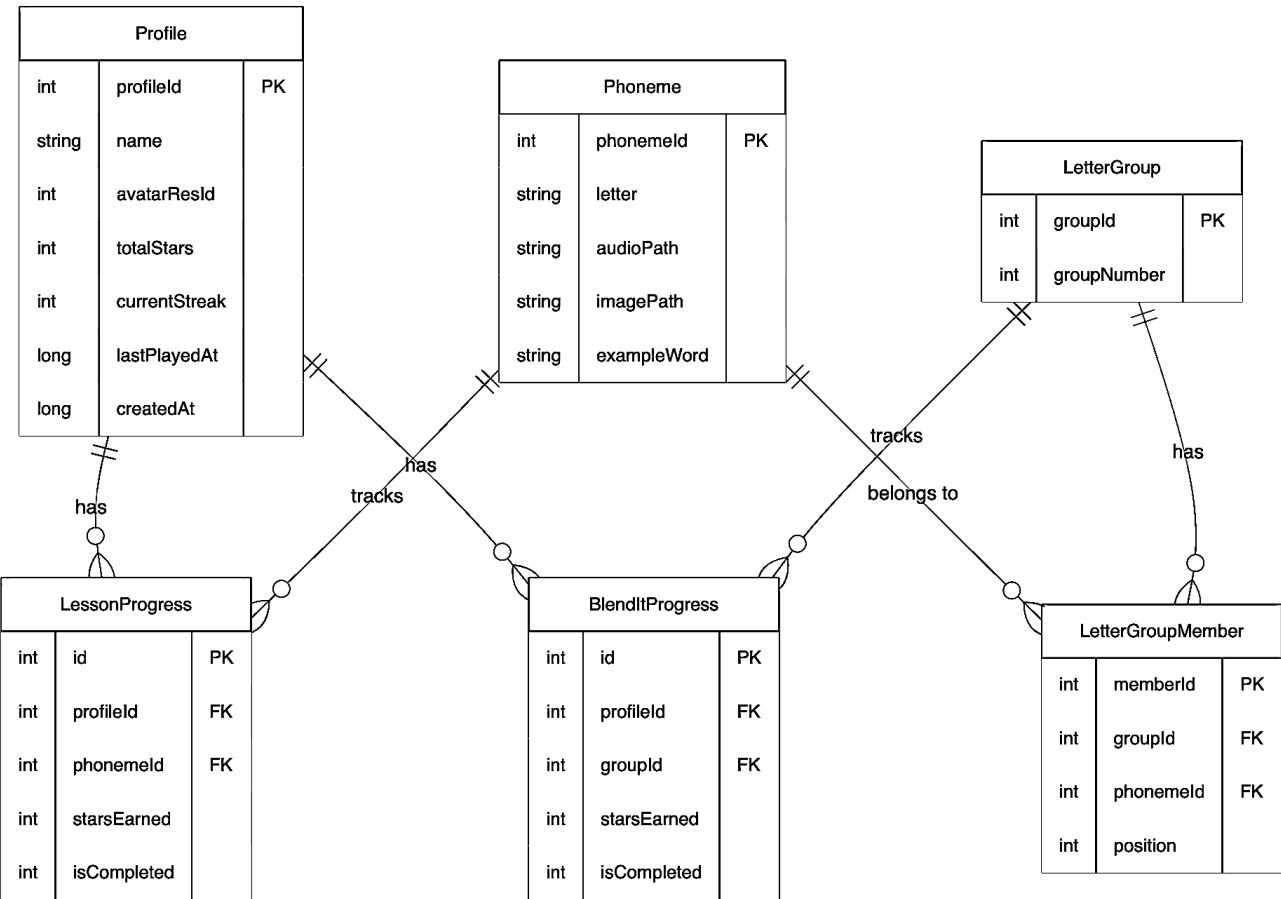
- Class Diagram



- Sequence Diagram



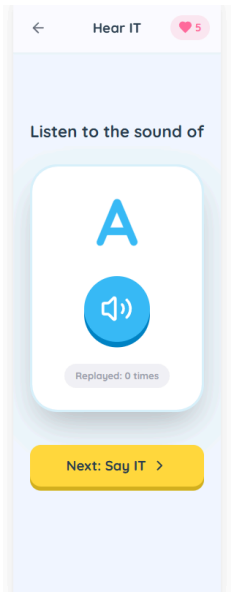
- Data Design



Module 2 Letter - Sound Acquisition (“Hear IT”)

2.1 Playback Phoneme Audio Model

- User Interface Design



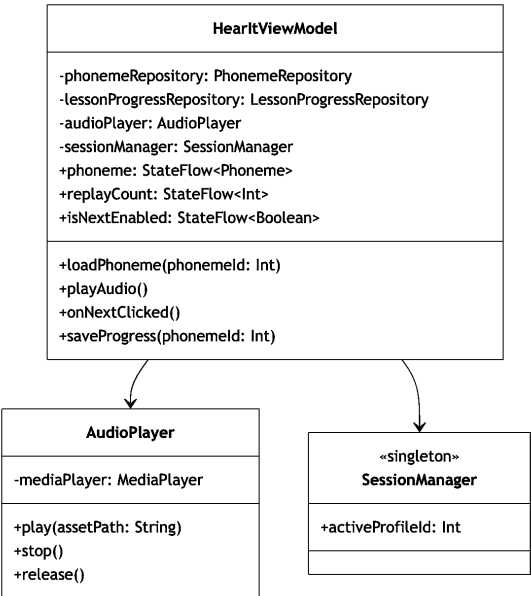
- **Front-end component(s)**

Component	Description	Type
HearItScreen	Root composable, loads phoneme data by phonemeld navigation argument	Screen
SubLevelProgressBar	3-step pill indicator showing Hear It as active step	Composable
AnimatedLetterCard	Displays letter symbol, phoneme notation, and example word picture	Composable
PlayButton	Large circular button that triggers phoneme audio playback	Composable
ReplayCounter	Dot indicators showing how many times audio has been replayed	Composable
MascotBubble	Instruction prompt from mascot — changes after first play	Composable
NextButton	Enabled only after child plays audio at least once	Composable

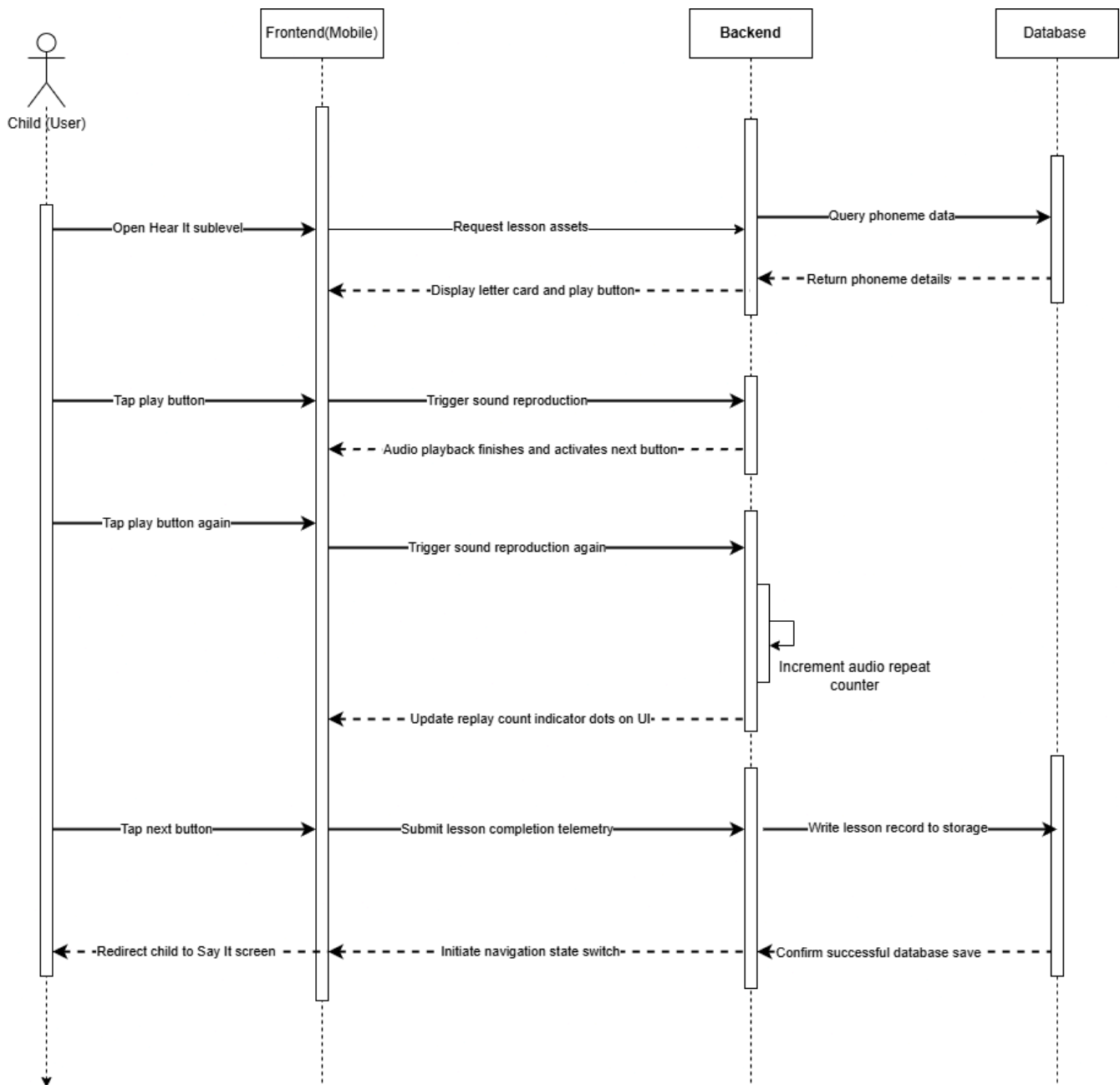
- **Back-end component(s)**

Component	Description	Type
HearItViewModel	Manages audio playback state, replay count, next button enablement, progress save	ViewModel
AudioPlayer	Loads and plays .mp3 files from assets folder	Class
PhonemeRepository	Provides phoneme data — letter, audio path, image path, example word	Interface
LessonProgressRepository	Saves Hear It completion status to LessonProgress for active profile	Interface
SessionManager	Provides activeProfileId for scoping progress writes	Object

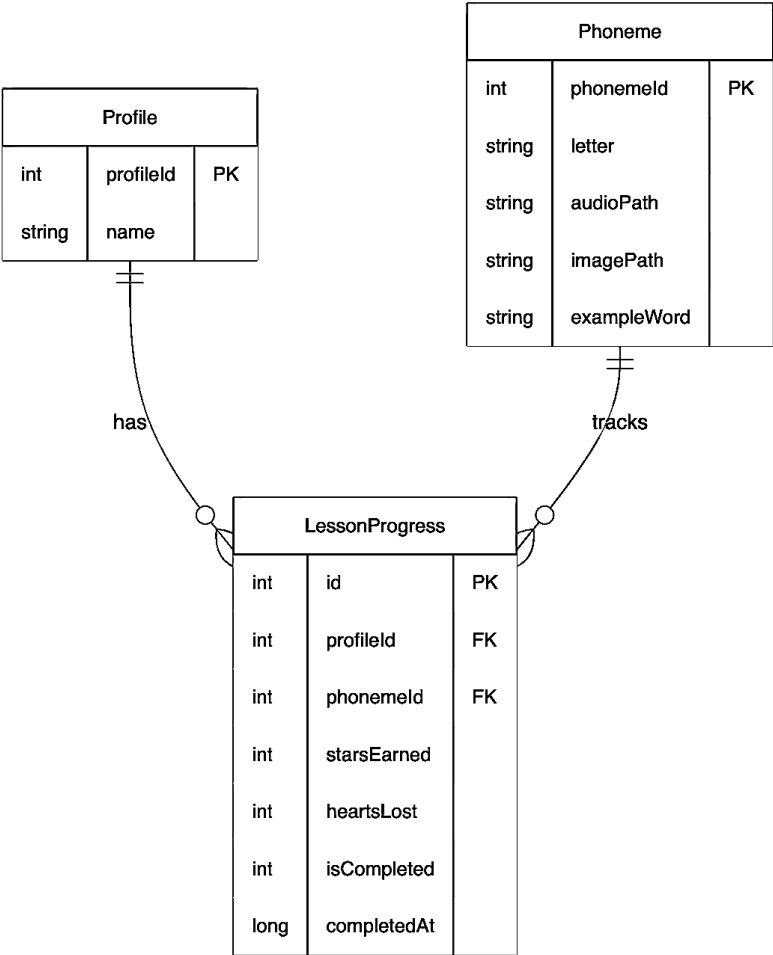
- **Object-Oriented Components**
 - **Class Diagram**



o Sequence Diagram



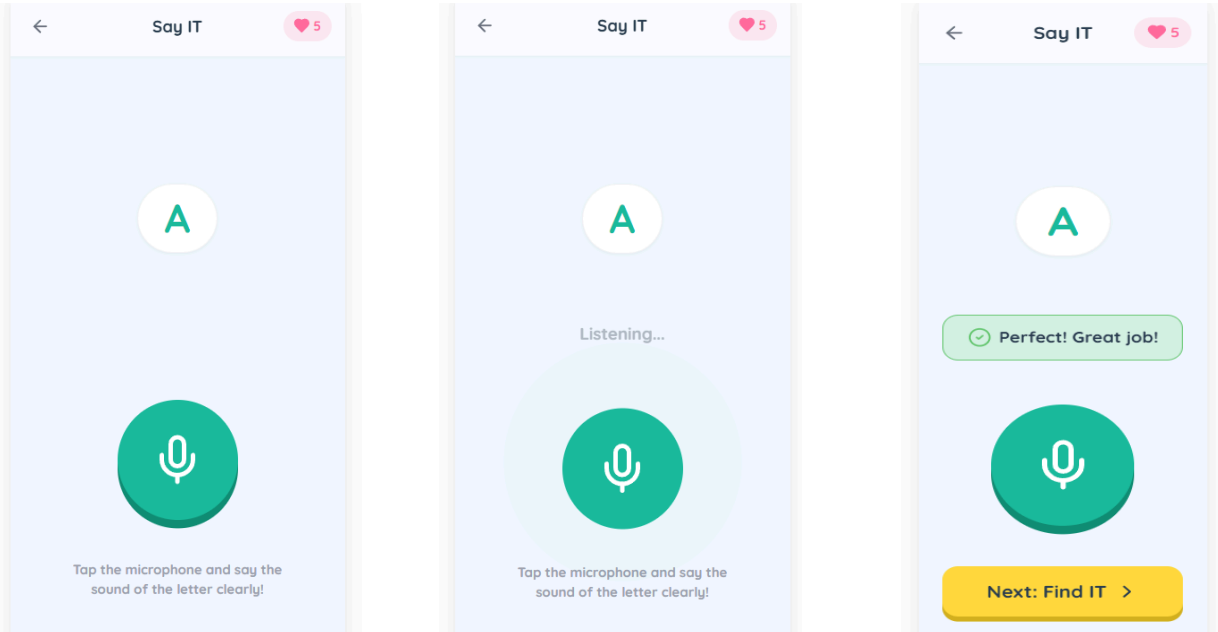
- Data Design



Module 3 Speech Production (“say IT”)

3.1 Process Speech Input and Provide Binary Feedback

- User Interface Design



- Front-end component(s)

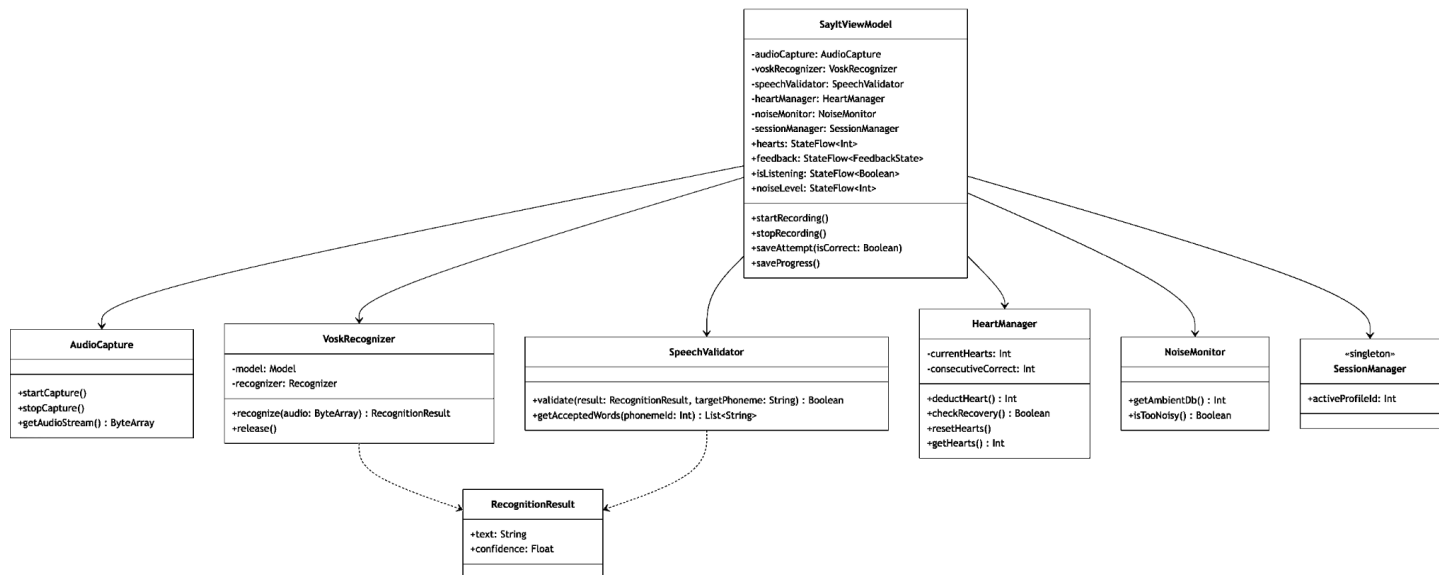
Component	Description	Type
SayItScreen	Root composable, loads phoneme by phonemeld navigation argument	Screen
SubLevelProgressBar	3-step pill indicator showing Say It as active step	Composable
HeartDisplay	5 hearts, animated on loss or recovery	Composable
PhonemePromptCard	Shows target letter and phoneme instruction text	Composable
MicrophoneButton	Hold-to-record button with pulse animation while held	Composable
ListeningAnimation	Waveform or pulse visual shown while microphone is active	Composable
FeedbackCard	Green or red binary result card shown after recognition with corrective audio	Composable
AttemptTracker	Dot row showing attempt history for current session, max 5 dots	Composable
NoiseLevelIndicator	Ambient dB bar — turns red and alerts child if noise exceeds 40dB	Composable
HeartRecoveryAnimation	+1 heart visual feedback when recovery triggers after 3 consecutive correct	Composable

- Back-end component(s)

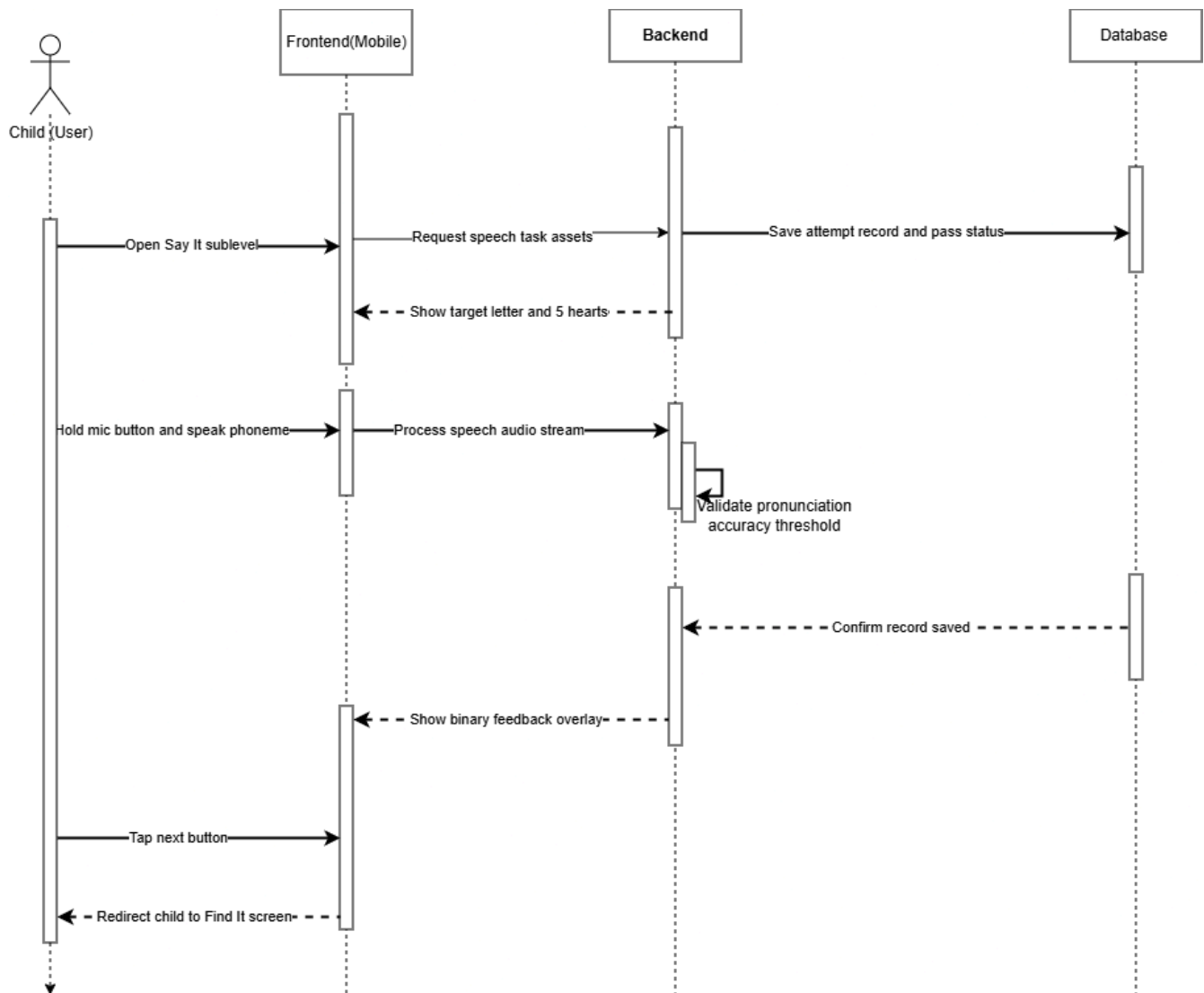
Component	Description	Type
SayItViewModel	Orchestrates recording, Vosk processing, heart logic, attempt logging, progress save	ViewModel
AudioCapture	Streams microphone input as PCM audio at 16kHz mono for Vosk	Class
VoskRecognizer	Wrapper for Vosk engine — returns recognized text and confidence score	Class
SpeechValidator	Checks recognized text against target phoneme accepted word list at 75% confidence threshold	Class

HeartManager	Manages heart count — deducts on fail, recovers per 3 consecutive correct, restarts at 3 on depletion	Class
NoiseMonitor	Reads ambient dB from microphone — alerts if above 40dB threshold	Class
AudioPlayer	Plays corrective phoneme audio on fail feedback	Class
SayItAttemptRepository	Saves each voice attempt with isCorrect and profileId to SayItAttempt	Interface
LessonProgressRepository	Saves Say It summary — isCompleted, heartsLost, starsEarned — for active profile	Interface
SessionManager	Provides activeProfileId for scoping all DB writes	Object

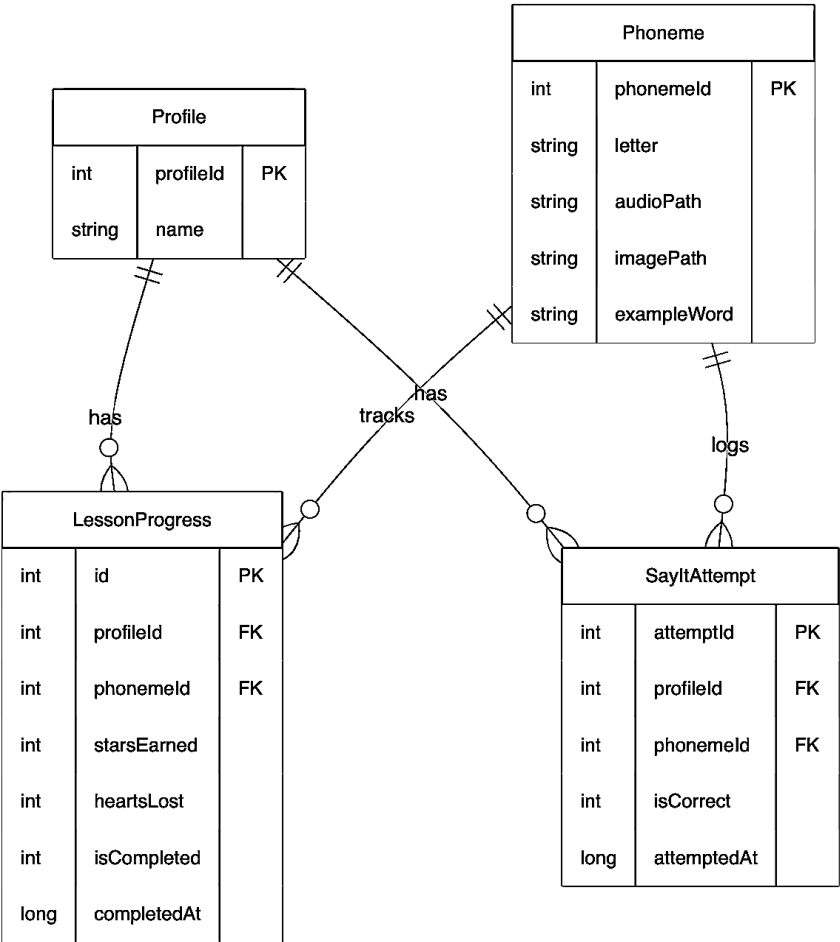
- Object-Oriented Components
 - Class Diagram



- Sequence Diagram



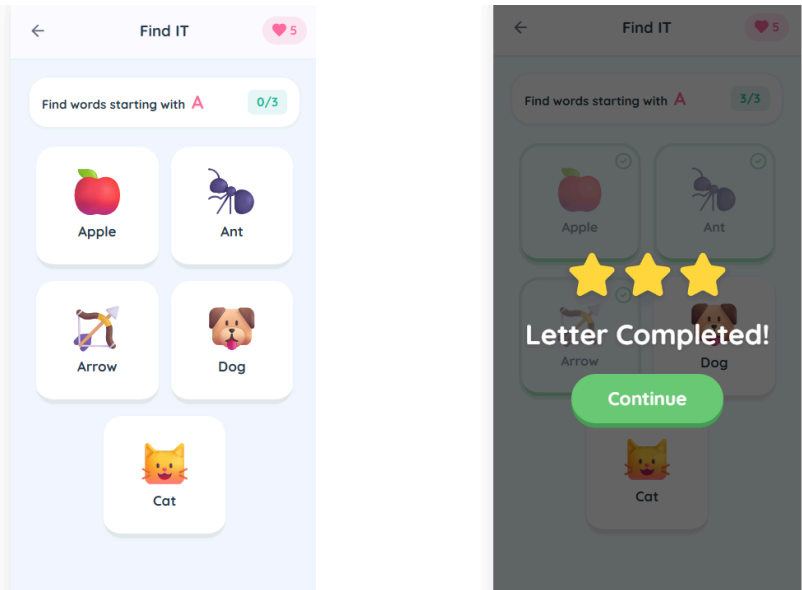
- Data Design



Module 4 Phoneme Discrimination (“find IT”)

4.1 Evaluate Dynamic Grid Selection

- User Interface Design



- Front-end component(s)

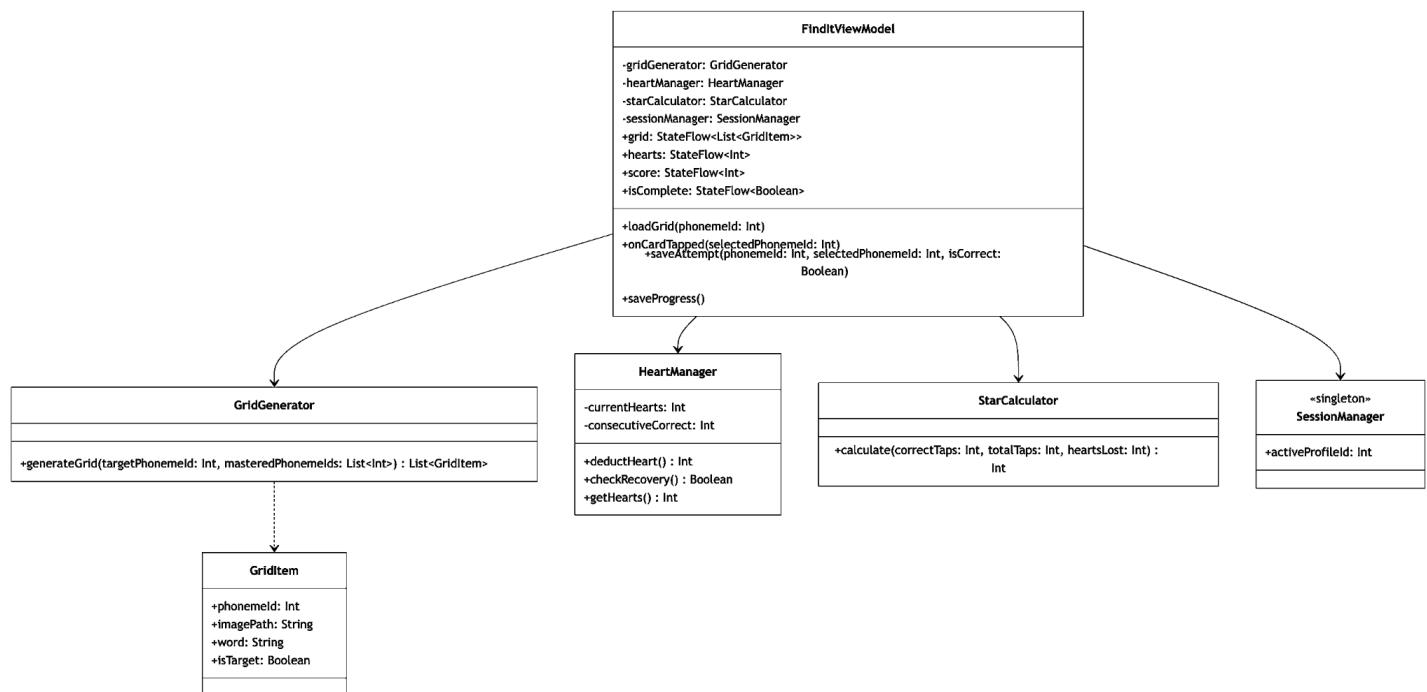
Component	Description	Type
FindItScreen	Root composable, loads phoneme and generates picture grid by phonemeld	Screen
SubLevelProgressBar	3-step pill indicator showing Find It as active step	Composable
HeartDisplay	Reused from Module 3 — same heart pool continues into Find It	Composable
ImageGrid	5-card responsive grid layout showing picture cards	Composable
PictureCard	Individual picture with word label — tap handling, green or red flash feedback	Composable
ScoreIndicator	Shows X of 3 targets found progress	Composable
MascotBubble	Starting instruction and mid-session encouragement prompts	Composable
CompletionAnimation	Celebration visual when all 3 target pictures are found	Composable
LetterCompleteScreen	Star drop-in animation and summary shown after all 3 sublevels complete	Screen
StarAnimation	1 to 3 stars dropping in with bounce animation	Composable

- Back-end component(s)

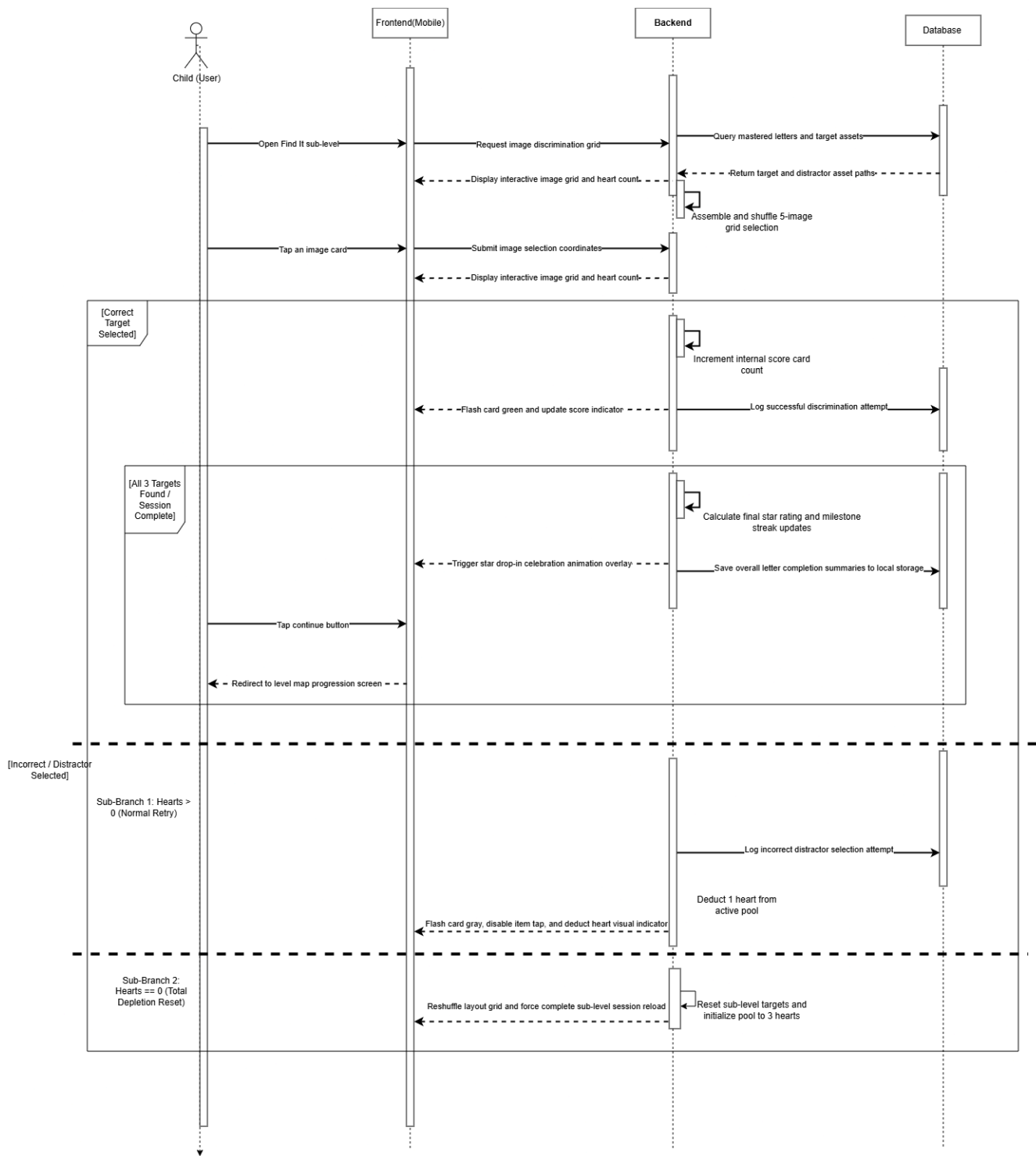
Component	Description	Type
FindItViewModel	Manages grid state, tap scoring, heart deductions, star calculation, attempt and progress save	ViewModel
GridGenerator	Selects 3 target images from current phoneme + 2 distractor images from mastered letters	Class
HeartManager	Reused from Module 3 — same instance continues, deducts on wrong tap	Class
StarCalculator	Calculates 1, 2, or 3 stars from accuracy and total hearts lost across Say It and Find It	Class
FindItAttemptRepository	Logs each picture tap with phonemeld,	Interface

	selectedPhonemeld, and isCorrect to FindItAttempt	
LessonProgressRepository	Saves Find It summary and final star rating to LessonProgress for active profile	Interface
PictureAssetRepository	Provides phoneme picture assets for grid generation	Interface
SessionManager	Provides activeProfileId for scoping all DB writes	Object

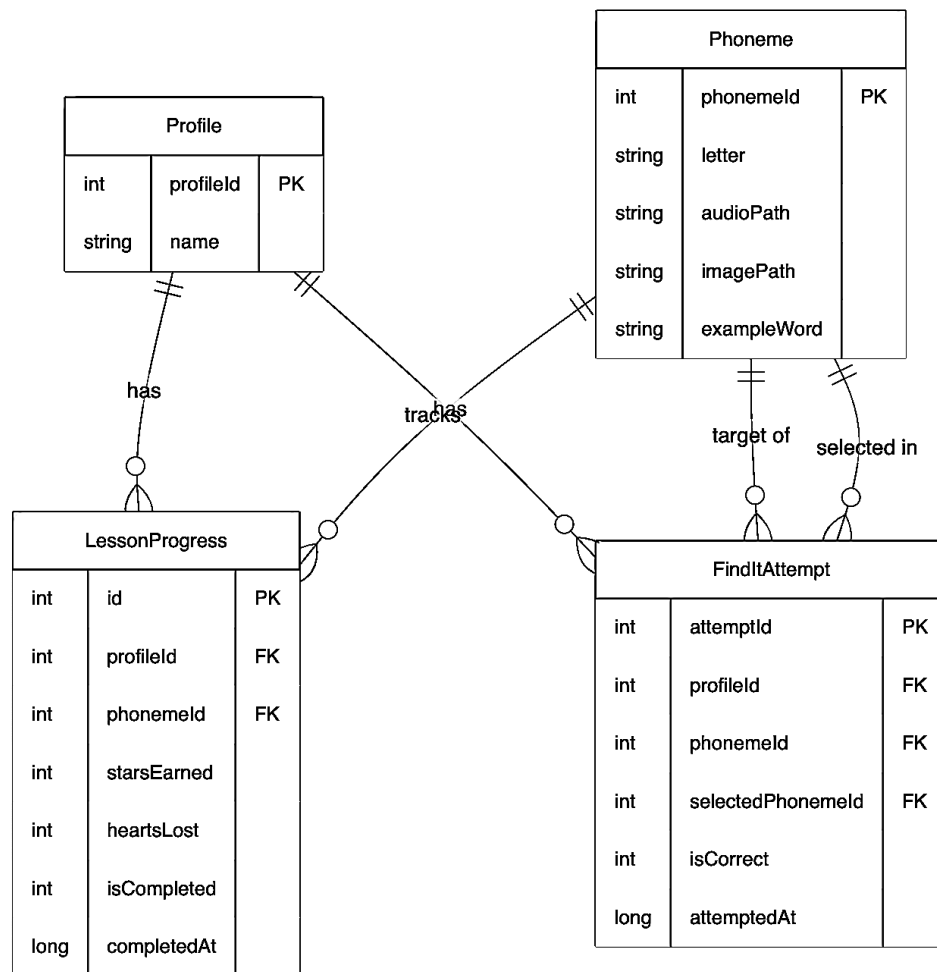
- Object-Oriented Components
 - Class Diagram



- Sequence Diagram



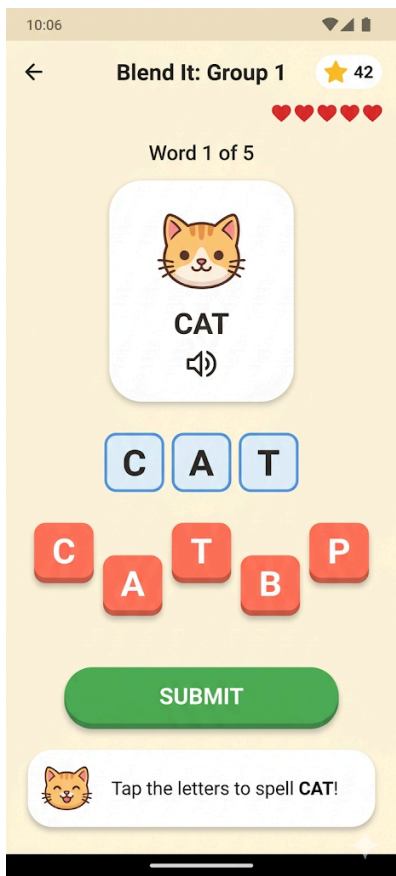
- Data Design



Module 5 Word Construction ("blendIT")

5.1 blend it

- User Interface Design



- Front-end component(s)

Component	Description	Type
BlendItScreen	Root composable, loads word session by groupId	Screen
BlendItCompleteScreen	Star drop-in animation + session summary after all words completed	Screen
TargetWordImage	Picture of the target word — tapping replays audio	Composable
WordAudioButton	Plays the target word audio automatically on load, replays on tap	Composable
LetterSlotRow	Row of empty slots matching the word length — fills as child taps tiles	Composable
LetterSlot	Individual drop target slot — shows letter when filled, tapping removes it back to bank	Composable

TileBank	Shuffled row of letter tiles at the bottom — correct letters + distractors	Composable
LetterTile	Individual tappable letter tile — grays out when placed in a slot	Composable
SubmitButton	Activates only when all slots are filled	Composable
WordFeedbackCard	Green/red result card shown after submission with corrective audio	Composable
HintIndicator	Shows how many hints have been used — triggers after 2 wrong attempts	Composable
HeartDisplay	5 hearts reset fresh per Blend It session, animated on loss	Composable
BlendItProgressIndicator	"X / 5 words completed" display	Composable
MascotBubble	Instruction and encouragement prompts	Composable
StarAnimation	1–3 stars dropping in with bounce animation on session complete	Composable

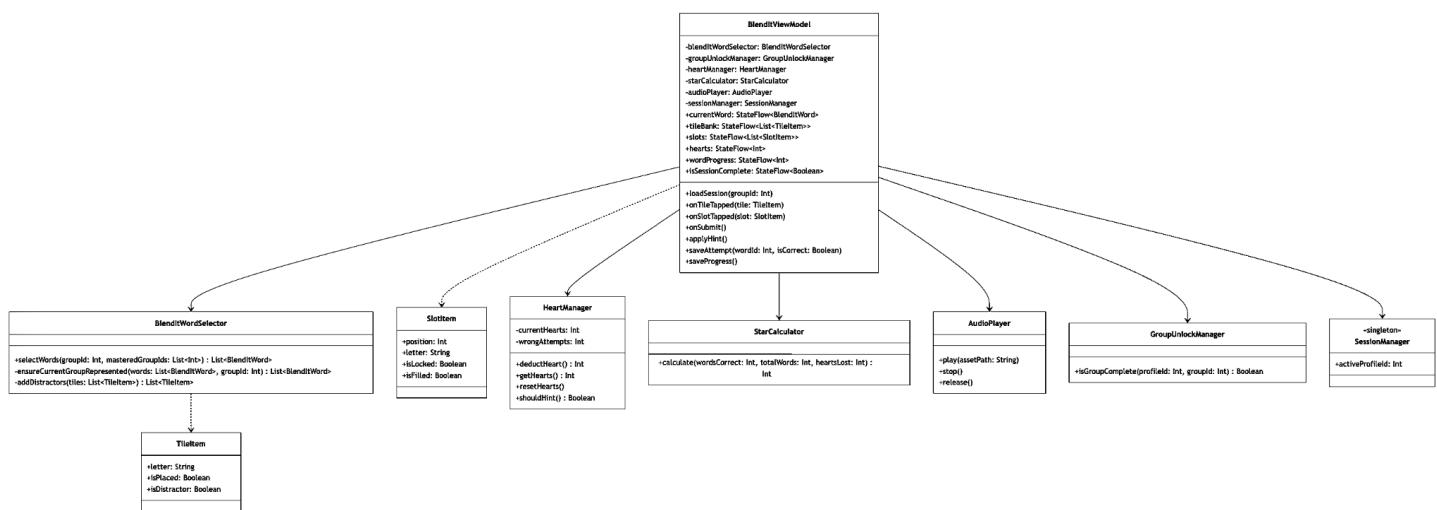
- **Back-end component(s)**

Component	Description	Type
BlendItViewModel	Orchestrates word session — manages tile state, slot state, hint logic, heart deductions, star calculation, DB save	ViewModel
BlendItWordSelector	Queries BlendItWord for all groups learned so far, ensures at least 1 word from current group, randomly selects 5	Class
GroupUnlockManager	Verifies all 4 letters in the group are fully complete in LessonProgress before allowing Blend It to load	Class
HeartManager	Fresh 5-heart pool per session — reused class from Module 3, new instance	Class
StarCalculator	Calculates 1/2/3 stars from words correct and hearts lost — reused from Module 4	Class
AudioPlayer	Plays word audio from assets on load and on replay tap — reused from Module 2	Class
SessionManager	Provides activeProfileId to scope all DB writes	Object

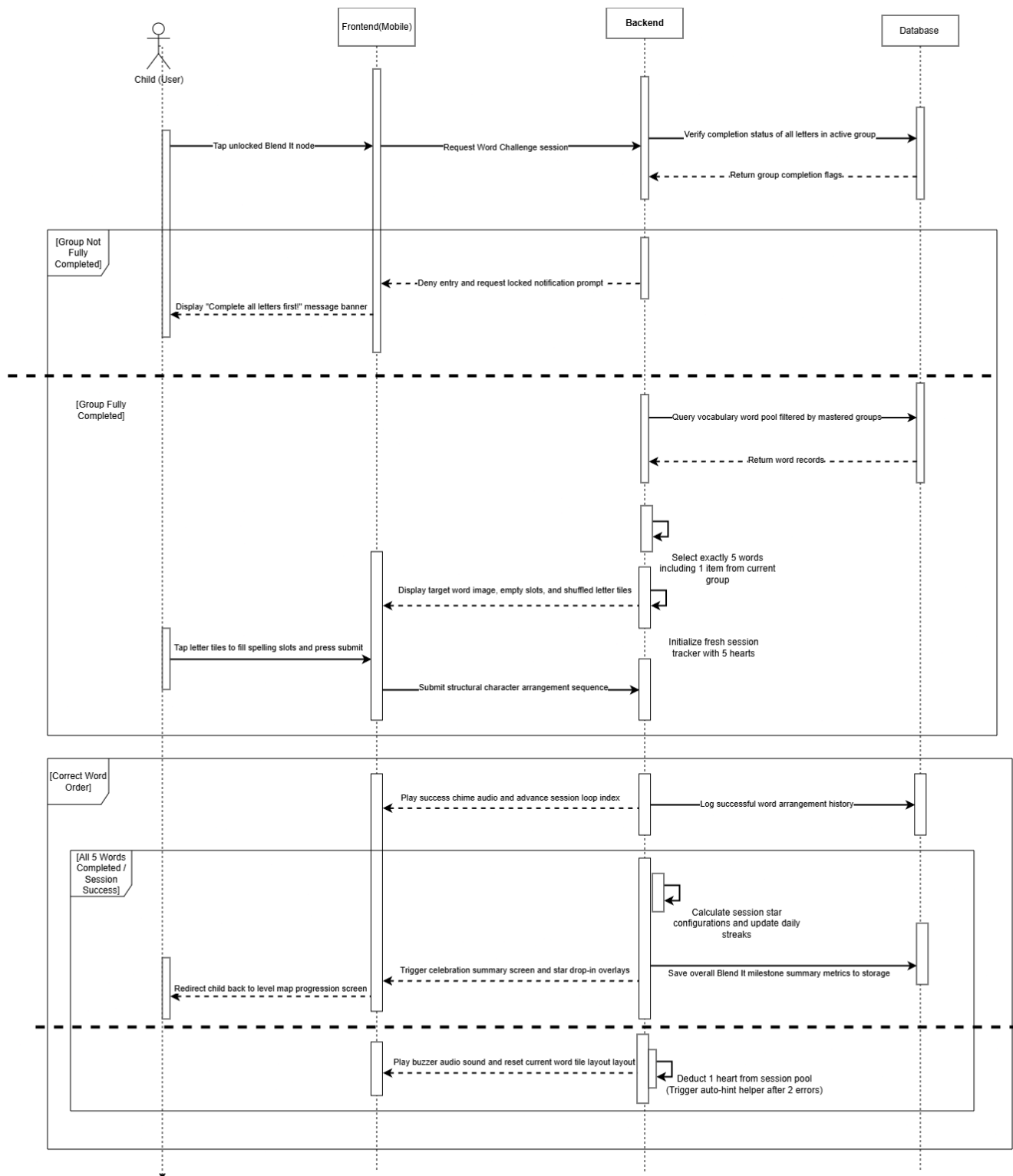
BlendItWordRepository	Provides word pool from BlendItWord table filtered by mastered groups	Interface
BlendItAttemptRepository	Logs each word attempt (wordId, isCorrect, profileId, groupId) to BlendItAttempt	Interface
BlendItProgressRepository	Saves session summary (starsEarned, heartsLost, isCompleted) to BlendItProgress per profile per group	Interface

- Object-Oriented Components

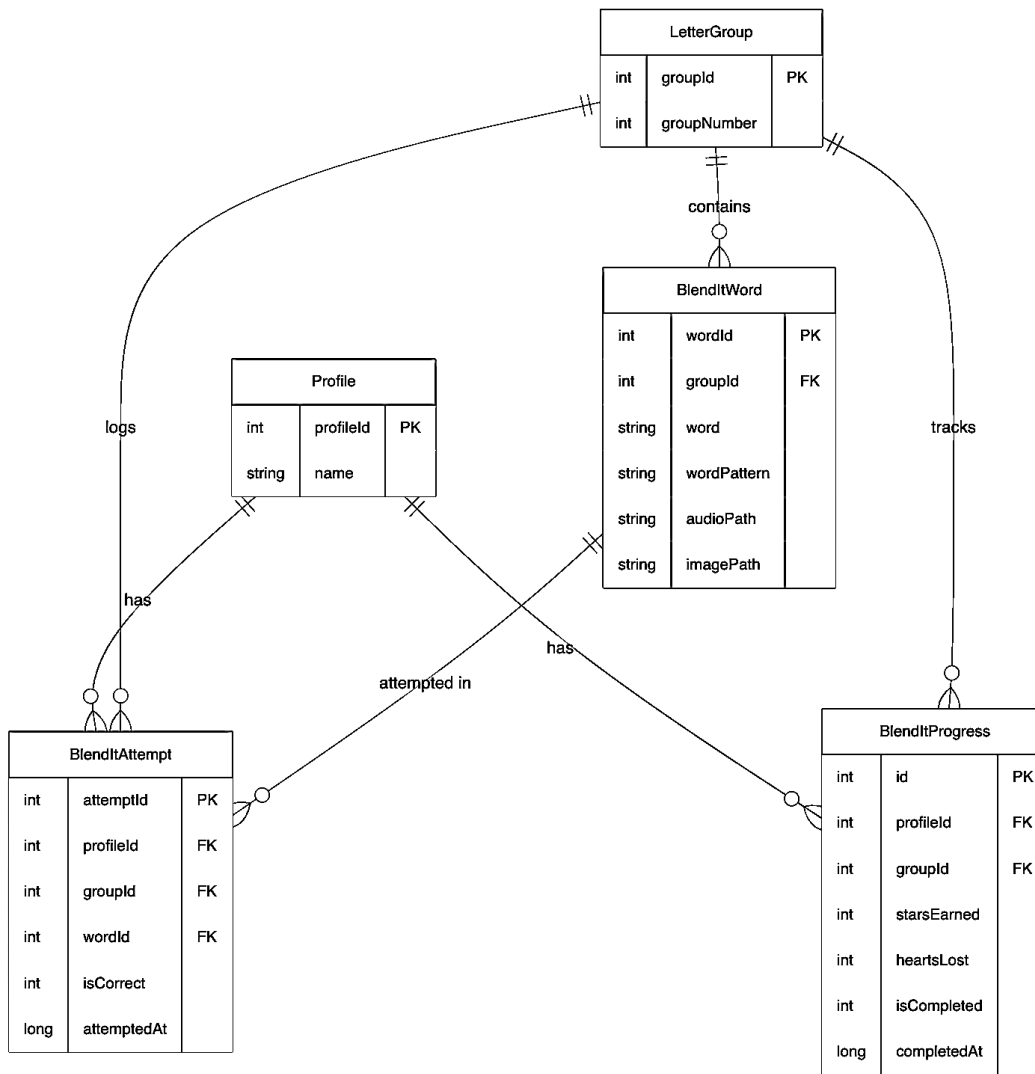
- Class diagram



- Sequence diagram



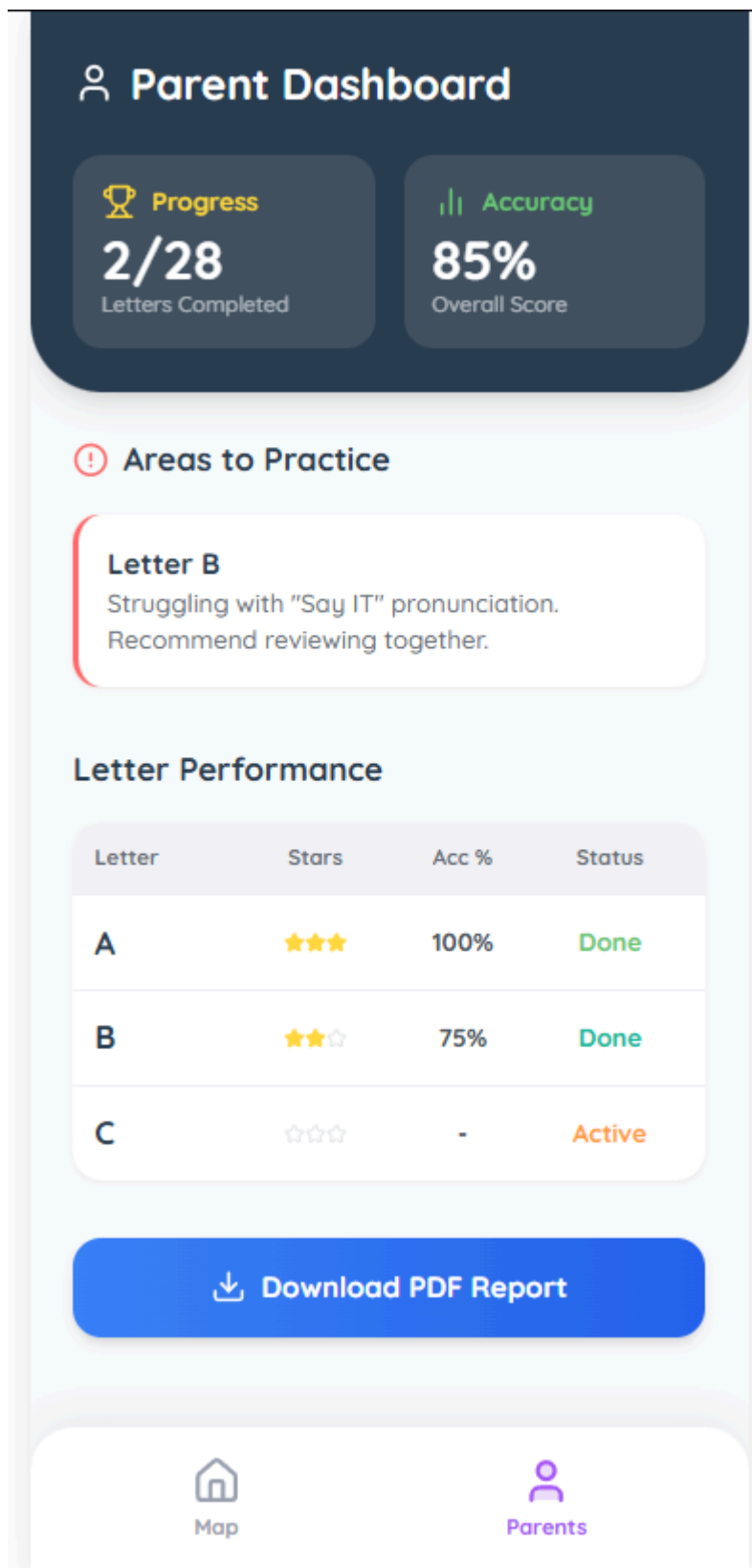
- Data design



Module 6 Parent Dashboard and Reporting

6.1 Present Interactive Progress Dashboard and PDF export report

- User Interface Design



- Front-end component(s)

Component	Description	Type
ParentDashboardScreen	Main dashboard — no login required, fully offline, profile-aware	Screen
ProfileSwitcherDropdown	Dropdown or tab at top of dashboard to switch between child profiles	Composable
OverallStatsCard	Summary card — completed letters, overall accuracy, current streak, total stars	Composable
LetterPerformanceTable	Scrollable 28-row table showing each letter with color-coded green yellow red status	Composable
BlendItSummaryCard	Shows Blend It stars earned per group and cumulative word accuracy	Composable
AtRiskSection	Highlighted list of red and yellow flagged letters needing attention	Composable
ExportButton	Triggers PDF generation for selected profile	Composable
ArithmeticGuardDialog	Simple addition popup shown before destructive actions like profile delete	Composable
ReportPreviewScreen	PDF content preview with save to local storage button	Screen

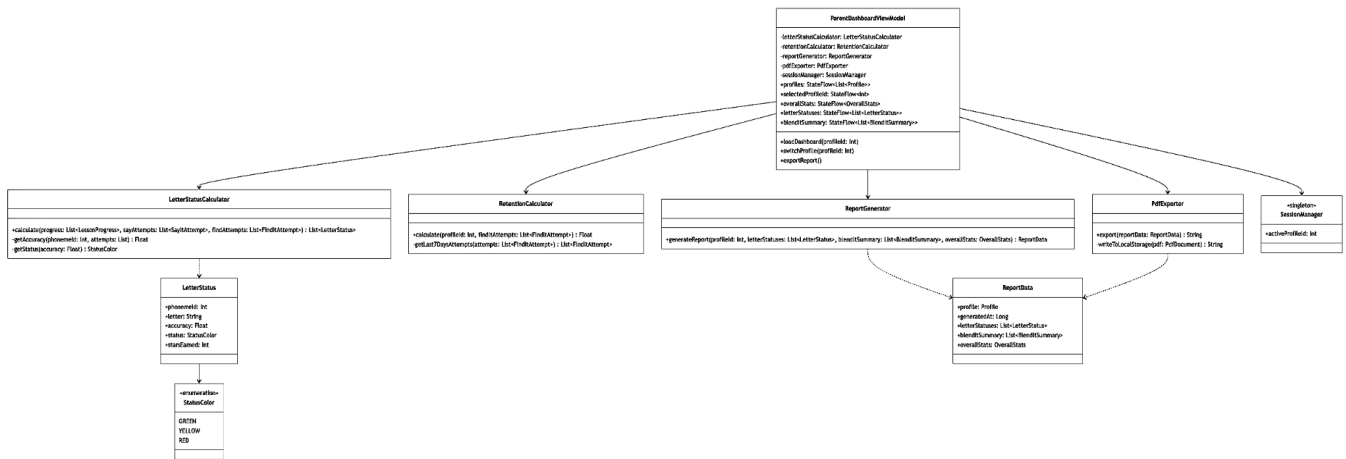
- Back-end component(s)

ParentDashboardViewModel	Aggregates all metrics from LessonProgress, SayItAttempt, FindItAttempt, BlendItProgress for selected profileId	ViewModel
ProfileRepository	Loads all profiles for the switcher dropdown	Interface
ProgressRepository	Queries LessonProgress, SayItAttempt, FindItAttempt for letter-level analytics	Interface
BlendItProgressRepository	Queries BlendItProgress and BlendItAttempt for word challenge analytics	Interface
LetterStatusCalculator	Applies green 80% or above, yellow 50 to 79%, red below 50% thresholds per letter	Class
RetentionCalculator	Calculates 7-day retention score from FindItAttempt distractor logs	Class

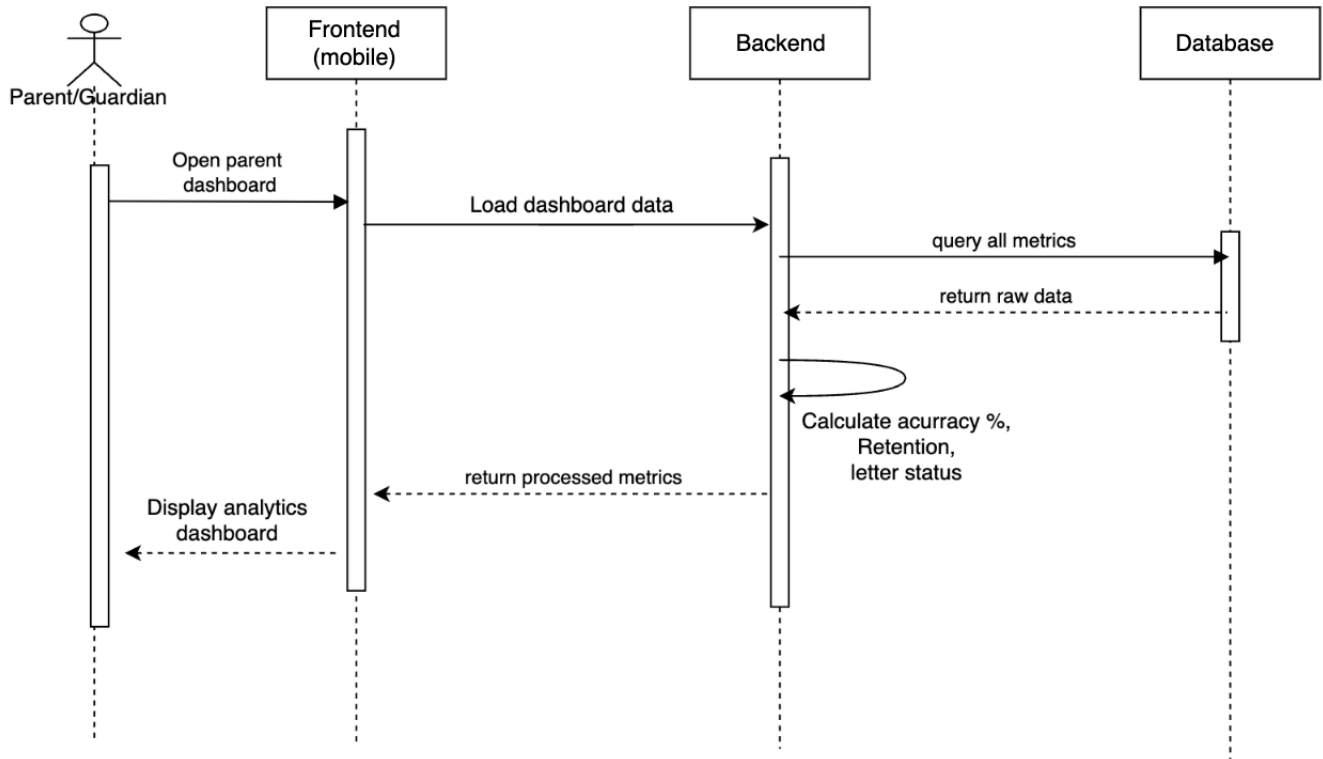
ReportGenerator	Compiles all metrics into a structured report data object for PDF export	Class
PdfExporter	Uses Android PdfDocument API to write formatted PDF to local storage	Class
ReportLogRepository	Saves file path and generation timestamp to ReportLog for selected profile	Interface
SessionManager	Provides activeProfileId as default selected profile on dashboard open	Object

- **Object-Oriented Components**

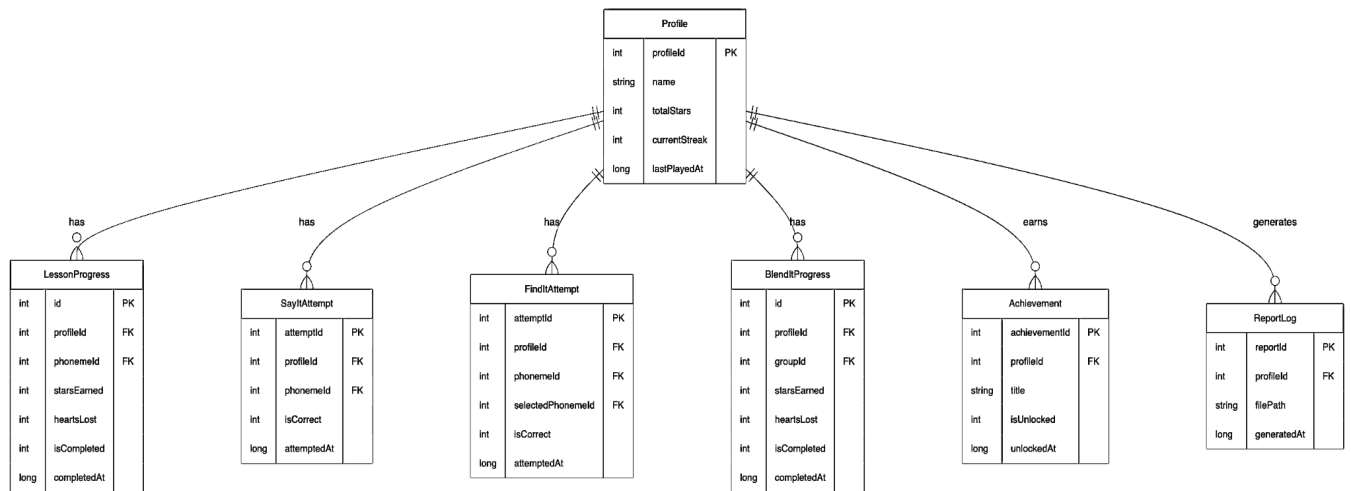
- **Class Diagram**



- **Sequence Diagram**



• Data Design



System ERD

